

Total No. of Questions: 22

Total No. of Pages: 02

Roll No. _____

Paper Code
3E1206

B.Tech. III-Sem (Main & Back) 2025-26
Aeronautical Engineering
3AN2-01 Advanced Engineering Mathematics-I
AN,AG,AE,CE,CR, EC,EI,ME,MH,PI,PT,BM
3E1206

Time : 3 Hours

Maximum Marks : 70

Attempt all ten questions from Part A, five questions out of seven questions from Part B and three questions out of five questions from Part C.

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. Units of quantities used / calculated must be stated clearly. Use of following supporting material is permitted during examination. (Mentioned in form No. 205)

1. _____ 2. _____

Part A

(Answer should be given up to 25 words only)

All questions are compulsory.

10x2=20

1. With usual notation prove that $\mu = E^{-\frac{1}{2}} + \frac{1}{2}\delta$
2. Write Newton cotes Quadrature formula.
3. ✓ Write the corrector formula of Adam's predictor corrector method.
4. Define transcendental equation with examples.
5. ✓ Find Laplace transform of $\cos^2 2t$.
6. Write convolution theorem for Inverse Laplace transform.
7. Write Fourier sine transform of $\frac{e^{-ax}}{x}$.
8. ✓ Write Fourier cosine transform of $f(t)$.
9. ✓ Find z-transform of $\cos h n\theta$.
10. ✓ Find inverse z-transform of $\frac{4z}{z-a}, |z| > |a|$

Part-B

(Analytical/Problem solving questions)

Attempt any five questions.

5x4=20

1. Use stirling formula to find Y_{28} , given $y_{20} = 49225, y_{25} = 48316, y_{30} = 47236, y_{35} = 45926, y_{40} = 44306$.
2. Find the root of $x^4 - x - 10 = 0$ which is nearer to $x=2$ by Newton Raphson method.
3. Solve $\frac{d^2x}{dt^2} + 9x = \cos 2t$, if $x(0) = 1, x\left(\frac{\pi}{2}\right) = -1$
4. ✓ Use Euler's method to solve $\frac{dy}{dx} = \frac{y^2 - x}{y^2 + x}$ given $y=1$ at $x=0$ also.
Find y for $x=0.1, 0.2$ and 0.3 .
5. ✓ Find Laplace transform of $\sin \sqrt{t}$
6. Find the Fourier transform of e^{-mx} hence show that
$$\int_0^{\infty} \frac{x \sin mx}{1+x^2} dx = \frac{\pi e^{-m}}{2}, m > 0$$
7. Find Z-transform of following - (i) $n^2 e^{an}$ (ii) $\sin(3n+5)$

Part-C

(Descriptive / Analytical / Problem Solving/ Design question)

Attempt any three questions.

3x10=30

1. ✓ Find value of $\int_1^2 \frac{1}{x} dx$ by (i) Simpson's 1/3 rule (ii) Trapezoidal rule
2. ✓ Using fourth order Runge-Kutta method with one step, compute $y(0.1)$ to five places of decimal, if
 $y' = 0.31 + 0.25y + 0.3t^2$ and $y = 0.72$ when $t = 0$
3. Apply convolution theorem to evaluate $L^{-1}\left[\frac{s^2}{(s^2 + a^2)(s^2 + b^2)}\right]$
4. An infinite string is initially at rest and that the initial displacement is $f(x) -\infty < x < \infty$. Determine the displacement $y(x,t)$ of the string.
5. Using Z-transform solve $U_{n+2} - 2U_{n+1} + U_n = 3n + 5$